Basic Structure and Process: Studies on Topology, Order, Formal Calculus, and Logic by the Lattice Topology Group at Sichuan University

Dedicated to the 130th Anniversary of Mathematics Disciplines at Sichuan University

Hui Kou, Maokang Luo*, Dexue Zhang

School of Mathematics, Sichuan University, Chengdu 610064, China

E-mail: kouhui@scu.edu.cn, makaluo@scu.edu.cn, dxzhang@scu.edu.cn

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**Corresponding author*

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Abstract

Since the 1960s, the interaction between general topology and other branches of mathematics (such as mathematical logic, algebra, and the theory of ordered structures) has become a prominent feature and constituted its primary research content. Under the leadership of Paoming Pu and Yingming Liu, the topology team at Sichuan University has achieved numerous internationally influential results in interdisciplinary fields concerning basic structures and processes in mathematics, including topology, ordered structure theory, formal calculus, and many-valued logic, establishing an academic tradition with distinctive characteristics and deep heritage. This paper briefly introduces a selection of work by the Lattice Topology Group at Sichuan University in two areas: lattice-valued topology and domain theory. The fundamental feature of these studies is an emphasis on the connection and integration of order, topology, category, and logic in terms of both structure and process.

Keywords: topology, order, domain theory, fuzzy topology, lattice-valued topology

MSC (2020) Subject Classifications: 06B35, 06F30, 18B35, 54A40, 54B30, 03B52

1. Introduction

Topology research at Sichuan University began with Mr. Paoming Pu. Pu was a pioneer of systolic geometry; in 1952, he published Pu's inequality for the real projective plane [95], which had a profound impact.[1] From 1952 to 1984, Pu served continuously as the Head of the Department of Mathematics at Sichuan University, making major contributions to the development of mathematics at Sichuan University, especially the initiation of topology research and the establishment of research teams. By the 1970s, the Department of Mathematics at Sichuan University had become an important base for topology, with a substantial research team whose representative figures included Paoming Pu, Yingming Liu, Jiguang Jiang, and Haoxuan Zhou.

In 1965, Zadeh [125] founded fuzzy set theory; in 1968, Chang [7] introduced fuzzy topological spaces. Supported by Mr. Zhaozhi Guan, the topology research group at Sichuan University quickly initiated research on fuzzy topology. In 1977, a joint paper by Paoming Pu and Yingming Liu proposed the key concept of the **quasi-coincident neighborhood of a fuzzy point**, extending the basic theory of general topology to fuzzy topological spaces; this received high praise and won the 4th Class National Natural Science Award in 1982.[2]

Shortly after Zadeh proposed fuzzy sets, Goguen [23, 24] published seminal works in 1967 and 1969 on fuzzy set theory and many-valued logic. The central thesis of these two papers was that complete lattices endowed with specific algebraic structures (such as complete residuated lattices) can and should replace the closed interval $[0, 1]$ as the logic truth-value domain. Consequently, lattice-valued fuzzy sets (L-fuzzy sets) became the primary objects of study in fuzzy set theory. In 1973, Goguen [25] introduced lattice-valued topological spaces and proved that whether Tychonoff's product theorem for covering compactness holds in lattice-valued topological spaces depends entirely on whether the top element of the lattice L is completely compact. Goguen's result revealed the natural connection between the order structure of a complete lattice L and lattice-valued topological properties, offering a new perspective for exploring the intrinsic relationship between order and topology.

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[1] *In his monograph [42], Katz of Bar-Ilan University, Israel, wrote: "Pu received his Ph.D. at Syracuse University in 1950 under the supervision of C. Loewner, resulting in the publication of the seminal paper [95] containing both inequalities (1.1.1) and (1.1.2)." ((1.1.2) refers to Pu's*

inequality).

[2] *The 1982 National Natural Science Award was the first evaluation resumed after its suspension between 1957 and 1981; the official title "National Natural Science Award" was also formally established at this time.*

The study of the relationship between order and topology has a long history: Stone's topological representation of Boolean algebras and distributive lattices in the 1930s [103–105]; Wallman's T_1 compactification in 1938 [108]; McKinsey and Tarski's work on topological algebra in 1944 [88]; and Nachbin's monograph on order and topology in 1965 [93]. The **domain theory** founded in the 1970s by Turing Award laureate Scott, mathematician Lawson, and others is the concentrated embodiment of research in this direction. Domain theory is closely tied to formal calculus, algebra, category theory, and logic, and plays a fundamental role in theoretical computer science; it has remained an active research field ever since its inception. In 1957, French mathematician Ehresmann [15] pointed out that complete Heyting algebras (complete lattices where finite meets distribute over arbitrary joins) can themselves be studied as generalized topological spaces. Such constructive investigations are deeply linked to intuitionistic logic, leading to the rich development of **point-free topology** (also known as Locale Theory; see Johnstone's 1982 monograph [38]).

Lattice-valued topology, domain theory, and point-free topology are distinct yet closely intertwined branches of mathematics. Core to all three is the connection and fusion of order and topological structures; theoretical breakthroughs in any single branch provide insights and foundations for the others. In 1986, Yingming Liu [69] coined the term **"Topology on Lattices" (格上拓扑学)**.[3] Following the evolutionary path of the field, "Topology on Lattices" can be understood as a general term encompassing research that investigates the intersection and integration of order and topological structures.

The topology team at Sichuan University has achieved prominent results in general topology, set-theoretic topology, lattice-valued topology, domain theory, and quantitative domain theory. Notable monographs include Jiguang Jiang's *Lectures on Selected Topics in General Topology* (1991) [37] and Yingming Liu & Maokang Luo's *Fuzzy Topology* (1997) [78]. Due to space limitations, this paper briefly surveys a portion of the team's work over the past forty years in **lattice-valued topology** and **domain theory**, characterized by the integration of order, topology, category, and logic across structural and process dimensions.

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[3] *A reviewer noted that the term "Topology on Lattices" appeared in Liu and He's 1985 paper [72]. In [72], the term appeared once without expansion. In [69], Liu clearly articulated that the concrete investigation combining order structures with topological structures possessed distinctive value.*

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2. Lattice-Valued Topology

In his 1965 seminal paper, Zadeh [125] pointed out that classes of objects in the real world often lack precisely defined criteria of membership.[4] Fuzzy sets serve as an effective tool for addressing these problems by grading membership, typically represented by a real number in $[0, 1]$.

Let X be a non-empty ordinary set. A fuzzy subset A of X is a mapping $A : X \to [0, 1]$, where $A(x)$ is interpreted as the degree of membership of x. When A is restricted to taking values in $\{0, 1\}$, it recovers the characteristic function of an ordinary subset. Since characteristic functions on X correspond bijectively to subsets of X, the concept of a fuzzy subset expands the codomain from the two-element Boolean algebra $\{0, 1\}$ to the unit interval $[0, 1]$. The unit interval $[0, 1]$ plays the role of a truth-value domain, extending classical two-valued logic to multi-valued (infinite-valued) logic. In 1967, Goguen [23] pointed out that complete lattices with specific algebraic structures (such as complete residuated lattices) can replace $[0, 1]$ as the logic truth-value domain.

Since the 1970s, fuzzy set theory has advanced along two primary tracks:

1. **Investigating internal laws of fuzziness:** Researching fuzzy logic and establishing mathematical theories to handle fuzzy phenomena.
2. **Developing mathematical theories based on fuzzy sets:** Establishing systems governed by multi-valued logic rules. Lattice-valued topology is one of its most successful developments, with Chinese scholars such as Yingming Liu and Guojun Wang making foundational contributions.

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[4] *"More often than not, the classes of objects encountered in the real physical world do not have precisely defined criteria of membership." — L. A. Zadeh, 1965.*

A **fuzzy topology** [7] on a set X is a family $\tau \subseteq [0,1]^X$ satisfying:

- (O1) $0_X, 1_X \in \tau$;
- (O2) $A, B \in \tau \implies A \wedge B \in \tau$;
- (O3) $\{A_i\}_{i \in I} \subseteq \tau \implies \bigvee_{i \in I} A_i \in \tau$.

The pair (X, τ) is called a fuzzy topological space, and elements of τ are open sets. If A is open, its complement $1 - A$ is a closed set.

Let (X, τ) and (Y, σ) be fuzzy topological spaces. A mapping $f : X \to Y$ is **continuous** if for every open set $B \in \sigma$, the preimage $f^{-1}(B) = B \circ f \in \tau$.

Fuzzy topology generalizes classical topology by replacing the powerset lattice 2^X with the complete lattice $[0,1]^X$ (pointwise order). However, under this definition, constant mappings between fuzzy topological spaces are not necessarily continuous. To resolve this, condition (O1) is strengthened to: *"τ contains all constant mappings $X \to [0,1]$"*, yielding a **stratified fuzzy topology**.

The category of stratified fuzzy topological spaces and continuous maps is denoted by $\mathbf{FTS}$. Given a topological space $(X, \mathcal{O})$, let $\omega(\mathcal{O})$ denote the family of all lower semicontinuous functions from $(X, \mathcal{O})$ to $[0,1]$. Then $\omega(\mathcal{O})$ is a stratified fuzzy topology on X, and $(X, \mathcal{O}) \mapsto (X, \omega(\mathcal{O}))$ defines a full and faithful functor:

$$\omega : \mathbf{Top} \to \mathbf{FTS}.$$

Thus, $\mathbf{Top}$ is isomorphic to a full subcategory of $\mathbf{FTS}$.

In fuzzy topology, the truth-value domain $[0,1]$ (or a complete lattice L) replaces $\{0,1\}$. Because $[0,1]$ possesses richer structure than $\{0,1\}$, fuzzy topology exhibits distinctive characteristics and technical machinery not present in classical topology.

2.1 Quasi-coincident Neighborhoods (重域)

A **fuzzy point** x_λ in X is a fuzzy set taking value $\lambda \in (0,1]$ at x and 0 elsewhere. A fuzzy point x_λ belongs to a fuzzy set A if $\lambda \leq A(x)$. In classical set theory, if a point belongs to a union of sets, it must belong to one of those sets (the *choice principle* [68–70]). However, for fuzzy points and fuzzy sets, this principle fails.

In 1977, Paoming Pu and Yingming Liu [96] introduced the **quasi-coincident relation (重于关系)**: a fuzzy point x_λ is **quasi-coincident** with a fuzzy set A ($x_\lambda \,\tilde{q}\, A$) if:

$$\lambda + A(x) - 1 > 0.$$

This relation restores the choice principle: $x_\lambda \,\tilde{q}\, \left(\bigvee A_i\right) \iff \exists i,\, x_\lambda \,\tilde{q}\, A_i$. If x_λ is quasi-coincident with the interior A°, then A is a **quasi-coincident neighborhood (Q-neighborhood)** of x_λ. The family of Q-neighborhoods of a fuzzy point forms a filter on $[0,1]^X$. Replacing standard neighborhood systems with Q-neighborhood systems, Pu and Liu successfully generalized Moore–Smith convergence theory to fuzzy topological spaces [71, 96, 97].

The Q-neighborhood theory was subsequently generalized to completely distributive lattices with order-reversing involutions [67]. For lattices without involutions, Guojun Wang [109, 110] introduced **remote neighborhood theory (远域理论)** as a dual counterpart using closed sets. In 1994, Mingsheng Ying [120] defined a fuzzy inclusion relation based on the Łukasiewicz implication:

$$\min\{1,\, 1 - \lambda + A(x)\},$$

and constructed a corresponding Moore–Smith convergence theory. Lowen [81], Lowen and Lowen [80], Höhle [31, 32], and others further developed minimal prime prefilters and fuzzy filter convergence theories [16, 18, 33, 35, 60, 61].

2.2 Lattice Structure and Topological Properties

Does altering the logical truth-value domain alter mathematical theorems? Since complete lattices do not generally satisfy the law of excluded middle, mathematical results depend heavily on the algebraic and order-theoretic properties of the truth-value lattice L.

Let L be a completely distributive lattice. For a topological space $(X, \mathcal{O})$, let $\omega_L(\mathcal{O})$ be the set of all lower semicontinuous maps from $(X, \mathcal{O})$ to L. This yields a full and faithful concrete functor:

$$\omega_L : \mathbf{Top} \to L\text{-}\mathbf{FTS}.$$

Spaces of the form $(X, \omega_L(\mathcal{O}))$ are called **induced spaces**. The functor ω_L has a concrete right adjoint ι_L and a concrete left adjoint ρ_L. In categorical terms [2], $\mathbf{Top}$ is isomorphic to a full subcategory of L-$\mathbf{FTS}$ that is simultaneously **concretely reflective and concretely coreflective**.

In their study of separation axioms in induced spaces [75], Yingming Liu and Maokang Luo established a lattice-valued version of the **Hahn–Dieudonné–Tong insertion theorem**:

> **Theorem 2.1 [76, 77]** *Let L be a completely distributive lattice equipped with the interval topology. The following are equivalent:*
>
> 1. *L has a countable strict join-generating set.*
> 2. *The L-valued Hahn–Dieudonné–Tong insertion theorem holds: for any normal space X, if $f : X \to L$ is lower semicontinuous, $g : X \to L$ is upper semicontinuous, and $g \le f$, there exists a continuous $h : X \to L$ such that $g \le h \le f$.*

> **Corollary 2.1 [76, 77]** *Let L be a completely distributive lattice with an order-reversing involution. If L has a countable strict join-generating set, then $(X, \mathcal{O})$ is normal if and only if the induced space $(X, \omega_L(\mathcal{O}))$ is normal in L-$\mathbf{FTS}$.*

The classical **Hewitt–Marczewski–Pondiczery theorem** states that if each space X_i has density $\le \kappa$ and $|I| \le 2^\kappa$, then the product space $\prod_{i\in I} X_i$ has density $\le \kappa$. Liu and Luo proved that its validity in L-topological spaces depends on whether the top element $1 \in L$ is coprime:

> **Theorem 2.2 [79]** *Let L be a completely distributive lattice with an order-reversing involution, and assume the top element 1 is coprime. Let $\{(X_i, \tau_i)\}_{i\in I}$ be a family of L-topological spaces and κ an infinite cardinal.*
>
> 1. *If each (X_i, τ_i) has density $\le \kappa$ and $|I| \le 2^\kappa$, then $\prod_{i\in I}(X_i, \tau_i)$ has density $\le \kappa$.*
> 2. *If each (X_i, τ_i) has density $\le \kappa$ and $|I| > 2^\kappa$, then $\prod_{i\in I}(X_i, \tau_i)$ has density $\le |I|$.*

> **Theorem 2.3 [79]** *Let L be a completely distributive lattice with an order-reversing involution. The following are equivalent:*
>
> 1. *The top element $1 \in L$ is coprime.*

2. *There exists a cardinal function f such that for any family of L-topological spaces with density $\leq \kappa$, their product has density $\leq f(\kappa, |I|)$.*

Connectedness of induced spaces is also characterized by the order properties of L:

> **Theorem 2.4 [79]** *Let L be a completely distributive lattice with an order-reversing involution. The following are equivalent:*
>
> 1. *L has no non-trivial complemented elements (no non-zero a, b with $a \vee b = 1$ and $a \wedge b = 0$).*
> 2. *Every L-topological space (X, τ) is connected if and only if the crisp topological space $(X, [\tau])$ is connected.*
> 3. *Every induced L-topological space $(X, \omega_L(\mathcal{O}))$ is connected if and only if $(X, \mathcal{O})$ is connected.*

For general complete lattices, Hongliang Lai and Dexue Zhang [50] established:

> **Theorem 2.5** *Let L be a meet-continuous complete lattice. The following are equivalent:*
>
> 1. *L is a continuous lattice.*
> 2. *$\mathbf{Top}$ is isomorphic to a full concretely reflective and concretely coreflective subcategory of L-$\mathbf{FTS}$.*

2.3 Connection with the Logical Structure of Real Numbers

The category $\mathbf{FTS}$ contains many full subcategories that are simultaneously concretely reflective and concretely coreflective; Lowen and Wuyts [85, 86] showed there are at least continuum-many such subcategories. Each forms a self-contained topological universe.

In 1982, Lowen [82] introduced **fuzzy neighborhood spaces ($\mathbf{FNS}$)**: a stratified space (X, τ) where there exists a family of filters $\{\mathcal{B}(x)\}_{x \in X}$ on $[0,1]^X$ such that for all $A \in [0,1]^X$:

$$\overline{A}(x) = \inf_{B \in \mathcal{B}(x)} \sup_{y \in X} \min\{A(y), B(y)\}.$$

$\mathbf{FNS}$ forms a concretely reflective and coreflective full subcategory of $\mathbf{FTS}$. In 2003, Sanjiang Li and Maokang Luo proved:

Theorem 2.6 [62] *The category of fuzzy neighborhood spaces* $\mathbf{FNS}$ *is not isomorphic to* $\mathbf{FTS}$.

A **continuous triangular norm (t-norm)** on $[0,1]$ is a continuous binary operation $*: [0,1] \times [0,1] \to [0,1]$ such that $([0,1], *, 1)$ is an abelian monoid and $*$ is monotone. It induces an implication operator:

$$x \to z = \sup\{y \mid x * y \leq z\}, \quad \text{satisfying} \quad x * y \leq z \iff y \leq x \to z.$$

The three basic continuous t-norms are:

- **Gödel t-norm:** $x * y = \min\{x, y\}$; $x \to y = 1$ if $x \leq y$, and y if $x > y$.
- **Product t-norm:** $x * y = xy$; $x \to y = 1$ if $x \leq y$, and y/x if $x > y$.
- **Łukasiewicz t-norm:** $x * y = \max\{0, x + y - 1\}$; $x \to y = \min\{1, 1 - x + y\}$.

Replacing the Gödel t-norm ($\min$) with a general continuous t-norm $*$ yields **$*$-fuzzy neighborhood spaces**. Lowen's original $\mathbf{FNS}$ corresponds to the Gödel t-norm; Höhle's probabilistic topologies [31, 32] correspond to the Łukasiewicz t-norm; and Lowen’s **approach spaces** [83, 84] correspond to the product t-norm.

In 1972, Scott [100] proved that injective T_0 topological spaces are precisely continuous lattices endowed with the Scott topology. Wei Li, Junche Yu, and Dexue Zhang demonstrated that this theorem depends heavily on the logical structure of the truth table:

Theorem 2.7 [63, 121] *Let $*$ be a continuous t-norm whose implication $\to$ is continuous off the diagonal $\{(x,x) \mid x \in [0,1]\}$. The following are equivalent:*

1. *The T_0-injective objects in the category of $*$-fuzzy neighborhood spaces are precisely the fuzzy continuous lattices equipped with the fuzzy topology generated by fuzzy Scott closed sets.*[5]
2. *The truth table $([0,1], *, 1)$ satisfies the double negation law: $(x \to 0) \to 0 = x$ for all $x \in [0,1]$ (i.e., $*$ is isomorphic to the Łukasiewicz t-norm).*

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[5] *Unlike the classical setting, because $([0,1], *, 1)$ does not generally satisfy the double negation law, fuzzy Scott open sets need not be complements of fuzzy Scott closed sets.*

For spaces determined by neighborhood systems of crisp points, Morsi, Hongliang Lai, and Dexue Zhang proved:

> **Theorem 2.8 [53, 91]** *Let $*$ be a continuous t-norm. The following are equivalent:*
>
> 1. *For any $a \in [0,1]$, if $*$ restricted to $[a^-, a^+]$ is isomorphic to the Łukasiewicz t-norm, then $a^- = 0$.*
> 2. *The implication $\to$ is continuous outside the diagonal.*
> 3. *The class of stratified fuzzy topological spaces determined by neighborhood systems of crisp points forms a concretely reflective and coreflective full subcategory of $\mathbf{FTS}$.*

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3. Domain Theory

Domain theory originated with Turing Award laureate Scott [99] in his study of denotational semantics for functional programming languages. In the 1960s, Strachey used the untyped λ-calculus to describe programming language semantics, but faced a fundamental obstacle: self-application $(\lambda x.x\,x)$ caused paradoxical circularities. In the late 1960s and early 1970s, Scott discovered that directed-complete partial orders (dcpos) carry an intrinsic topology—the **Scott topology**—which allowed the construction of semantic "domain models" for the λ-calculus, placing denotational semantics on solid mathematical foundations [99–102, 106].

Concurrently, pure mathematicians led by Lawson uncovered continuous lattices while studying compact topological semilattices [56–58]. These discoveries attracted both pure mathematicians and theoretical computer scientists, rapidly developing domain theory and leading to major monographs [1, 4, 20, 21, 26].[6]

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[6] *Broadly, "domain theory" refers to semantic domains for programming languages, including continuous posets, algebraic dcpos, and continuous lattices [1, 4, 26].*

A poset P is a **directed-complete poset (dcpo)** if every directed subset $D \subseteq P$ has a supremum $\bigvee D$.

> **Definition 3.1 [1]** *Let P be a poset and $x, y \in P$. We say x is **way below** y ($x \ll y$) if for every directed subset $D \subseteq P$, $y \leq \bigvee D$ implies $x \leq d$ for some $d \in D$. An element satisfying $k \ll k$ is called **compact**. Write $\Downarrow x = \{y \in P \mid y \ll x\}$ and $k(x) = \{k \in P \mid k \ll k \leq x\}$.*
>
> 1. *P is a **continuous domain** if it is a dcpo and for all $x \in P$, $\Downarrow x$ is directed with $x = \bigvee \Downarrow x$.*
> 2. *P is an **algebraic domain** if it is a dcpo and for all $x \in P$, $k(x)$ is directed with $x = \bigvee k(x)$.*
> 3. *A continuous (resp. algebraic) domain that is a complete lattice is called a **continuous lattice** (resp. **algebraic lattice**).*

A subset $U \subseteq P$ is **Scott open** if $U = \uparrow U$ and for every directed $D \subseteq P$, $\bigvee D \in U \implies U \cap D \neq \emptyset$. The Scott topology is denoted $\sigma(P)$. While $(P, \sigma(P))$ is typically only T_0, Lawson [56, 58] introduced the **Lawson topology** (the join of the Scott topology and the lower topology), which is Hausdorff for continuous domains. A continuous domain that is compact in its Lawson topology is called a **Lawson-compact continuous domain**.

A mapping $f : P \to Q$ between dcpos is **Scott continuous** if and only if it is monotone and preserves directed suprema. The category of dcpos with Scott-continuous maps is denoted $\mathbf{DCPO}$.

Chinese research on domain theory began in the 1980s. In 1986, Yingming Liu [69] highlighted Dongsheng Zhao's 1985 work on when Scott spaces are sober [127]. In 1995, Yixiang Chen [8] completed the first domestic Ph.D. thesis on domain theory. Concurrently, Jihua Liang and Xiaoquan Xu produced significant results [64–66, 74, 116]. In 1999, the 1st International Symposium on Domain Theory and Its Applications (ISDT 1999) was held at Shanghai Normal University.[8] Over forty years, Chinese researchers have contributed across stable domains [9, 10], sobriety and well-filteredness [112, 115, 119], convergence theory [111, 130], domain environments [36, 59, 113, 128], and probabilistic powerdomains [27]. In 2022, Xiaoquan Xu published the monograph *Order and Topology* [117].

The Sichuan University team was among the earliest domestic groups in domain theory, making contributions to foundational structures [45, 46, 64], function spaces [44, 47, 48, 65, 74], powerdomains [19, 87, 122–124], and Cartesian closed categories [36, 114, 129].

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[8] *The ISDT conference series has been held ten times and continues to the present.*

3.1 Function Spaces on Ordered Structures

A topological space X is **sober** if every irreducible closed set is the closure of a unique point. X is **core-compact** if its topology $\mathcal{O}(X)$ is a continuous lattice. Continuous domains endowed with the Scott topology are always locally compact and sober.

For topological spaces X and Y, the set of continuous maps $[X \to Y]$ can be equipped with topologies such as the topology of pointwise convergence, the compact-open topology, and the **Isbell topology** $\mathrm{Is}(X, Y)$, generated by subbasic sets:

$$N(H, V) = \{f \in [X \to Y] \mid f^{-1}(V) \in H\}, \quad (H \in \sigma(\mathcal{O}(X)),\, V \in \mathcal{O}(Y)).$$

When X is locally compact and sober, the Isbell topology equals the compact-open topology.

If Y is a dcpo with the Scott topology, $[X \to Y]$ ordered pointwise is also a dcpo, and carries its own Scott topology. In 1990, Lawson and Mislove raised two open questions in *Open Problems in Topology* [107]:

1. *Characterize dcpos L such that $[X \to L]$ is a continuous domain for every core-compact space X. If X is restricted to compact core-compact spaces, does the class of dcpos remain the same?*
2. *Under what conditions on a continuous dcpo L does the Scott topology on $[X \to L]$ coincide with the Isbell topology for all core-compact spaces X?*

Yingming Liu and Jihua Liang [74] resolved both questions in 1996 using **Jung's classification theorems** for maximal Cartesian closed categories of continuous domains:

> **Theorem 3.1 (Jung [40, 41])**
>
> 1. *For continuous domains E and P, if $[E \to P]$ is a continuous domain, then E is Lawson-compact or P is an L-domain (continuous local lattice).*
> 2. *The categories $\mathbf{CL}$ (L-domains) and $\mathbf{FS}$ (finite-separation domains) are the only two maximal Cartesian closed full subcategories of $\mathbf{CONT}$ (pointed continuous domains).*

> **Theorem 3.2 (Liu & Liang [74])** *For any dcpo L, the following are equivalent:*
>
> 1. *L is a continuous L-domain.*
> 2. *$[X \to L]$ is a continuous L-domain for every core-compact space X.*
> 3. *$[X \to L]$ is a continuous domain for every core-compact space X.*
>
> *However, there exists a dcpo T that is not an L-domain such that $[X \to T]$ is continuous for every compact core-compact space X (completely solving Lawson–Mislove Problem 532).*

> **Theorem 3.3 (Liu & Liang [74])** *For any pointed continuous L-domain P, the following are equivalent:*
>
> 1. *The Scott topology on $[X \to P]$ coincides with the Isbell topology for every core-compact space X.*
> 2. *P is a bounded-complete domain (partially solving Lawson–Mislove Problem 535).*

For continuous L-domains, Maokang Luo and Hui Kou [47] established:

> **Theorem 3.4 [47]** *If P is a continuous L-domain, the Scott topology on $[P \to P]$ equals the Isbell topology if and only if P is Lawson-compact.*

Regarding Lawson–Mislove Problem 544 on characterizing pairs (X, L) where $[X \to L]$ is Lawson-compact, Liang and Keimel [44] proved:

> **Theorem 3.5 [44]** *A dcpo L is a Lawson-compact continuous L-domain if and only if $[X \to L]$ is a Lawson-compact continuous domain for every core-compact space X with Property W.*

Luo and Kou [48] introduced the strictly weaker **Property RW**:

> **Theorem 3.6 [48]**
>
> 1. *Every Lawson-compact continuous domain has Property RW. For continuous L-domains, Lawson compactness is equivalent to the Scott topology having Property RW.*
> 2. *For a pointed dcpo P, $[X \to P]$ is a Lawson-compact continuous domain for all core-compact spaces X with Property RW if and only if P is a Lawson-compact continuous L-domain.*

> 3. *For a core-compact space X, $[X \to P]$ is Lawson-compact for all pointed Lawson-compact continuous L-domains P if and only if X has Property RW.*

3.2 Cartesian Closed Categories

In a Cartesian closed category (CCC), exponential objects satisfy:

$$[(X \times Y) \to Z] \cong [X \to [Y \to Z]],$$

providing semantics for higher-order function types and the λ-calculus.

While the maximal Cartesian closed subcategories of $\mathbf{CONT}$ are classified ($\mathbf{CL}$ and $\mathbf{FS}$), a major open question emerged following Jones and Plotkin's [39] introduction of the **probabilistic powerdomain**: *Is the category of $\mathbf{FS}$-domains closed under the probabilistic powerdomain?* (The **Jung–Tix problem**).

To overcome these constraints, researchers explored generalizations:

- **Quasi-continuous domains:** Introduced by Gierz et al. [22] in 1983. Kou et al. [46] showed that a quasi-continuous domain is continuous if and only if it is **meet-continuous**. Using this, Jia, Jung, Kou, et al. [36] established:

> **Theorem 3.7 [36]** *If P is a quasi-continuous domain and $[P \to P]$ is quasi-continuous, then P is meet-continuous, and hence continuous. Consequently, the category of quasi-continuous domains contains no Cartesian closed full subcategories beyond those already in $\mathbf{CONT}$.*

- **Stable domain theory:** Haoran Zhao and Hui Kou [129] identified a Cartesian closed category using stable functions and constructed the first universal domain in the sense of Scott:

> **Theorem 3.8 [129]** *Let $\mathbf{CCDI}_s$ denote the category of conditionally complete ω-DI-domains with stable maps.*
>
> 1. *$\mathbf{CCDI}_s$ is Cartesian closed.*
> 2. *$[T^\omega \to_s T^\omega]$ is a universal domain for $\mathbf{CCDI}_s$.*

- **Equilogical spaces:** Bauer, Birkedal, and Scott [6] introduced equilogical spaces (T_0 spaces with equivalence relations), forming a Cartesian closed category. Battenfeld, Schröder, and Simpson [5] showed that quotients of second-countable T_0 spaces form a complete, cocomplete CCC.
- **Directed spaces ($\mathbf{DTOP}$):** Yuxu Chen, Hui Kou, and Zhenchao Lyu [11–14] investigated directed spaces, establishing:

> **Theorem 3.9 [11–14]** *The category of directed spaces $\mathbf{DTOP}$ satisfies:*
>
> 1. $\mathbf{DTOP}$ *is a coreflective full subcategory of* $\mathbf{TOP}_0$.
> 2. $\mathbf{DCPO} \subset \mathbf{POSET} \subset \mathbf{DTOP} \subset \mathbf{TOP}_0$.
> 3. $\mathbf{DTOP}$ *is Cartesian closed and closed under Hoare, Smyth, and Plotkin power spaces and free topological cones.*

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4. Conclusion

In December 1978, at the Modern Mathematics Planning and Collaboration Symposium in Shanghai, Paoming Pu outlined new directions in general topology,[9] emphasizing its integration with other mathematical disciplines. Under the leadership of Paoming Pu and Yingming Liu, the Sichuan University topology group established a tradition of research across topology, mathematical logic, order structures, and many-valued logic. Since China's reform and opening-up, Sichuan University has served as a central hub for topology in China, training generations of scholars and contributing fundamentally to the development of the discipline.

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[9] *These directions included: (1) unification of topological structures; (2) studying spaces via mappings; (3) generalized metric space theory; (4) mathematical logic and general topology; (5) non-standard topology; (6) fuzzy topology; (7) bitopological spaces; (8) shape theory [90].*

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